

End-to-End Registry Migration & Reconciliation Framework

ETL, Validation & Holdings Reconstruction

1. Scenario

A new client is being onboarded into a registry platform environment.

The client provides historical registry extracts from a source registry system in .xlsx format.

The objective is to:

- ingest raw source extracts
 - validate ingestion integrity
 - transform source structures into target production schemas
 - validate transformed production structures against staging data
 - reconstruct downstream shareholder holdings
 - reconcile final migrated state against client source-of-truth holdings
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2. Source Data Files

File	Purpose
Company Static Data	Security / company master information
Shareholder Static Data	Holder master records
Transactions	Historical share registry movement ledger
Corporate Actions	Dividend and stock split events
Shareholder Holdings Snapshot	Final downstream source-of-truth holdings

3. Key Migration Concepts

Share Registry Holdings Reconstruction

Historical transactions and corporate actions are processed to reconstruct downstream shareholder holdings.

The transaction ledger includes:

- IPO issuance
- market transfers
- DRP unit issuance
- stock split unit generation

Corporate actions include:

- dividend events
- DRP participation
- stock splits

4. Migration Architecture

Client XLSX Extracts

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CSV Standardization

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Raw SQL Staging Ingestion

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Stage 1 Integrity Validation

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ETL for Target Schema

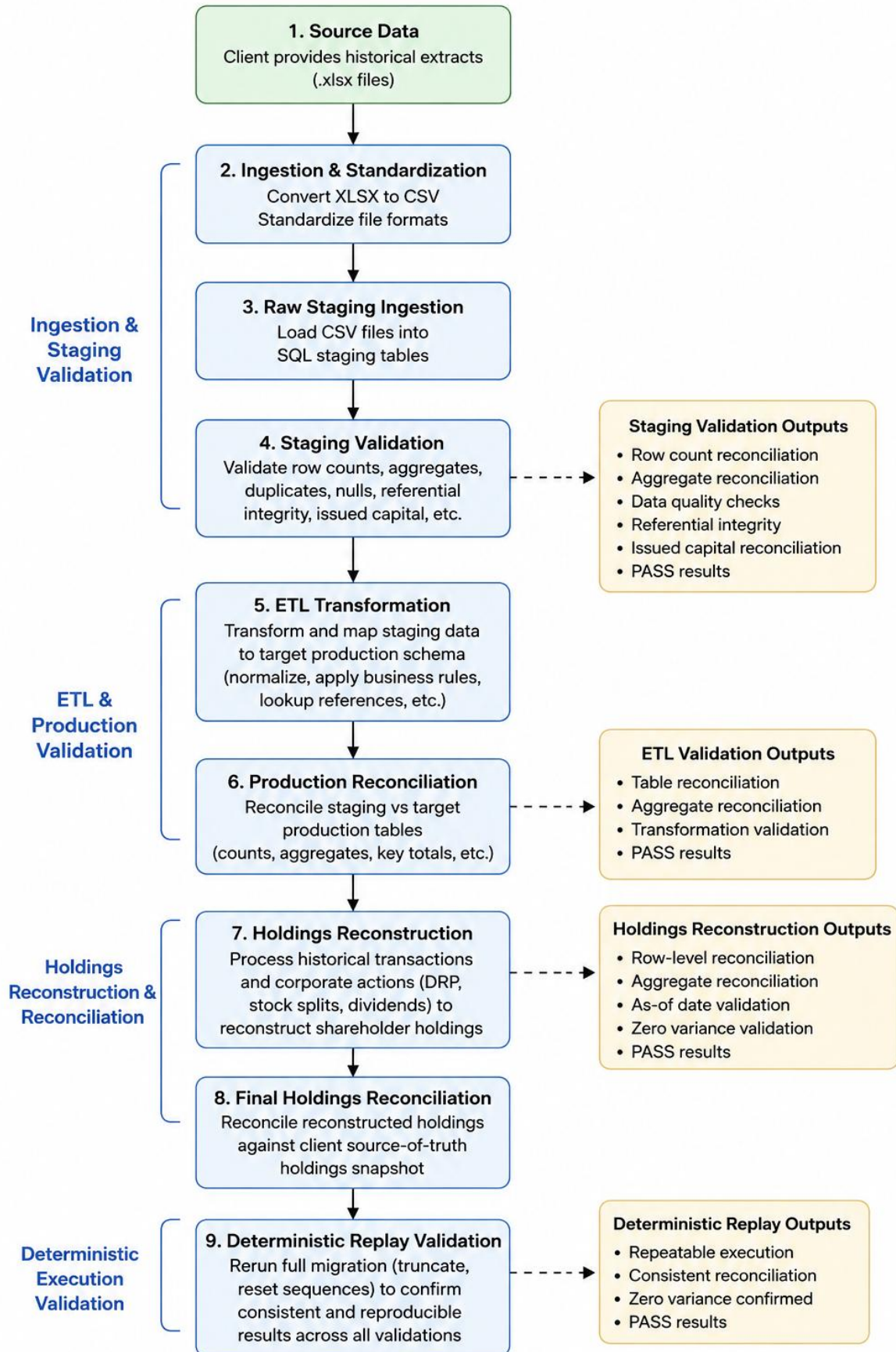
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Production Reconciliation

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Final Holdings Reconciliation

End-to-End Migration & Reconciliation Workflow



5. Migration Execution & Validation Evidence

The following sections demonstrate staging validation, ETL reconciliation controls, holdings reconstruction, and deterministic execution validation.

5.1.1 Staging Integrity Summary

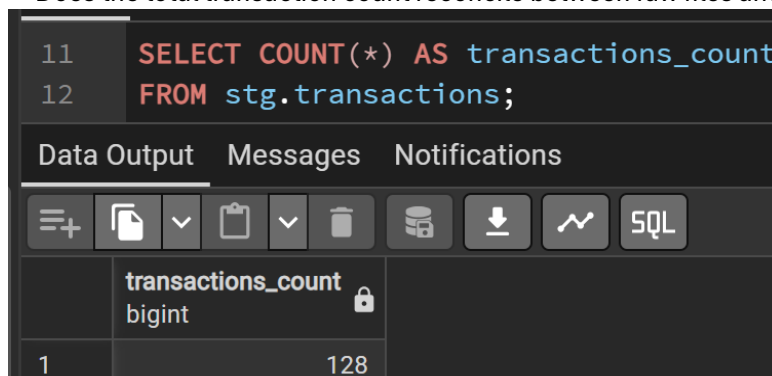
The following validation controls were executed between raw source extracts and SQL staging tables following ingestion.

Data Type	CSV	SQL	Variance	Status
<i>Company Data Row Count</i>	1	1	0	PASS
<i>Shareholders Row Count</i>	100	100	0	PASS
<i>Corporate Actions Row Count</i>	3	3	0	PASS
<i>Transactions Row Count</i>	128	128	0	PASS
<i>Transaction Aggregate Validation</i>	1535000	1535000	0	PASS
<i>Distinct Holder Validation</i>	32	32	0	PASS
<i>Transaction ID Null Validation</i>	0	0	0	PASS
<i>Holder Null Validation</i>	0	0	0	PASS
<i>Duplicate Transaction Validation</i>	0	0	0	PASS
<i>Issued Capital Reconciliation</i>	1535000	1535000	0	PASS
<i>Referential Integrity Validation</i>	0	0	0	PASS
<i>Holdings Position Validation</i>	32	32	0	PASS

5.1.2 Sample Validation Queries

1. Row Count Validation

Does the total transaction count reconcile between raw files and the staging table?



The screenshot shows a SQL query editor with the following query:

```
11 SELECT COUNT(*) AS transactions_count
12 FROM stg.transactions;
```

Below the query editor, there are tabs for "Data Output", "Messages", and "Notifications". The "Data Output" tab is active, showing a table with one row and one column:

	transactions_count bigint
1	128

2. Aggregate Validation

Does the sum of all transaction units match between raw files and staging table?

14	-- Aggregate Validation
15	SELECT
16	SUM(units) AS total_transaction_units
17	FROM stg.transactions;
Data Output Messages Notifications	
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	total_transaction_units numeric
1	1535000.0000

3. Duplicate Validation

Do any duplicate transaction IDs exist within the staging table?

37	-- Duplicate Transaction Validation
38	SELECT
39	txn_id,
40	COUNT(*) AS duplicate_count
41	FROM stg.transactions
42	GROUP BY txn_id
43	HAVING COUNT(*) > 1;
Data Output Messages Notifications	
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	txn_id character varying (20)
	duplicate_count bigint

4. Issued Capital Reconciliation

Does total issued capital reconcile against the aggregate transaction ledger?

45	-- Shares Issued Reconciliation
46	SELECT
47	SUM(t.units) AS reconstructed_units,
48	c.shares_outstanding,
49	SUM(t.units) - c.shares_outstanding AS variance
50	FROM stg.company_data c
51	CROSS JOIN stg.transactions t
52	GROUP BY c.shares_outstanding;
Data Output Messages Notifications	
≡ + 📄 ▼ 📋 ▼ 🗑️ 🗄️ ⬇️ 📈 SQL Showing rows: 1 to	
	reconstructed_units numeric
	shares_outstanding numeric (20)
	variance numeric
1	1535000.0000
	1535000
	0.0000

5. Referential Integrity Validation

Do all transaction holders reconcile against the shareholder master table?

54 **SELECT**

55 **COUNT(*) AS orphan_transaction_holders**

56 **FROM** stg.transactions t

57 **LEFT JOIN** stg.holder_master h

58 **ON** t.holder = h.holder_name

59 **WHERE** h.holder_name **IS NULL**;

Data Output Messages Notifications

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SQL

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orphan_transaction_holders

bigint

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Stage 1 validation confirms successful ingestion integrity between raw client extracts and SQL staging structures prior to ETL transformation processing.

5.2.1 Target Production Schema

Target Table	Purpose
target.company_master	Production company master
target.shareholder_master	Production shareholder registry
target.share_registry_transactions	Normalized registry transaction ledger
target.dividend_events	Dividend event processing
target.stock_split_events	Stock split processing
target.shareholder_ca_elections	DRP participation tracking

5.2.2 Mapping & Transformation Specification

Transformation mappings were implemented between raw staging structures and normalized production target schemas.

Company Master Transformation Mapping

Source Field	Target Field	Transformation Logic
company_name	company_name	Direct mapping
exchange	exchange_code	Production exchange normalization
currency	currency_id	Reference normalization via ref.currency

status	company_status	ACTIVE → A, otherwise → I
shares_outstanding	shares_outstanding	Direct numeric mapping

Shareholder Master Transformation Mapping

Source Field	Target Field	Transformation Logic
holder_name	shareholder_name	Direct mapping
holder_type	shareholder_type	INDIVIDUAL → I, FUND → F, TRUST → T, etc.
residency_country	residency_country_id	Reference normalization via ref.country
currency	currency_id	Reference normalization via ref.currency
holder_status	shareholder_status	ACTIVE → A, CLOSED → C, etc.
communication_preference	communication_preference	EMAIL → E, POST → P
holder_category	holder_category	RETAIL → R, INSTITUTIONAL → I, FOUNDER → F

Share Registry Transactions Transformation Mapping

Source Field	Target Field	Transformation Logic
holder	shareholder_id	Relationship reconstruction via join to shareholder master
txn_id	source_transaction_id	Direct source transaction preservation
txn_type	transaction_type	BUY → B, SELL → S, ISSUE → I, DRP → D, SPLIT → SP
units	units	Direct numeric mapping
txn_date	transaction_date	Direct date mapping
description	transaction_description	Direct mapping

5.2.3 ETL Validation Evidence

ETL Domain	Validation Objective	Validation Status
Company Master	Do transformed company attributes reconcile correctly between staging and target structures following ETL execution?	PASS
Shareholder Master	Do transformed shareholder records and aggregate counts reconcile between staging and target structures?	PASS
Corporate Actions	Do transformed corporate action records reconcile correctly following event decomposition and DRP normalization?	PASS

Transaction Ledger	Do transformed transaction records, aggregate counts, and unit balances reconcile between staging and production ledgers?	PASS
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1. Does total shareholder count reconcile between staging and target structures?

```

1  --Aggregate validation
2  SELECT
3      (SELECT COUNT(*) FROM stg.holder_master) AS stg_holder_count,
4      (SELECT COUNT(*) FROM target.shareholder_master) AS target_holder_count,
5      (
6          SELECT COUNT(*)
7          FROM stg.holder_master s
8          INNER JOIN target.shareholder_master t
9              ON s.holder_name = t.shareholder_name
10         ) AS matched_holder_count;

```

stg_holder_count bigint	target_holder_count bigint	matched_holder_count bigint
100	100	100

2. Do shareholder types reconcile correctly following ETL normalization?

```

27  CASE
28      WHEN s.holder_type = 'INDIVIDUAL'
29          AND t.shareholder_type = 'I'
30      THEN 'PASS'
31
32      WHEN s.holder_type = 'TRUST'
33          AND t.shareholder_type = 'T'
34      THEN 'PASS'
35
36      WHEN s.holder_type = 'FUND'
37          AND t.shareholder_type = 'F'
38      THEN 'PASS'
39
40      WHEN s.holder_type = 'JOINT'
41          AND t.shareholder_type = 'J'
42      THEN 'PASS'

```

holder_type character varying (50)	shareholder_type character varying (1)	holder_type_validation text
JOINT	J	PASS
JOINT	J	PASS
JOINT	J	PASS
SMSF	U	PASS

3. Do decomposed dividend and stock split events reconcile correctly following ETL transformation?

Query	Query History
1	--Aggregate decomposition
2	SELECT
3	(SELECT COUNT(*)
4	FROM stg.corporate_actions
5	WHERE UPPER(action_type) = 'DIVIDEND') AS stg_dividend_count,
6	
7	(SELECT COUNT(*)
8	FROM target.dividend_events) AS target_dividend_count,
9	
10	(SELECT COUNT(*)
11	FROM stg.corporate_actions
12	WHERE UPPER(action_type) = 'SPLIT') AS stg_split_count,
13	
14	(SELECT COUNT(*)
15	FROM target.stock_split_events) AS target_split_count;

Data Output	Messages	Notifications
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	stg_dividend_count bigint	target_dividend_count bigint
1	1	1

4. Has DRP participation been retained correctly following ETL normalization?

83	--DRP elections validation
84	SELECT
85	(SELECT COUNT(*)
86	FROM stg.holder_master
87	WHERE UPPER(drp_flag) = 'Y') AS stg_drp_yes_count,
88	
89	(SELECT COUNT(*)
90	FROM target.shareholder_ca_elections
91	WHERE election_type = 'DRP'
92	AND election_status = 'Y') AS target_drp_yes_count;

Data Output	Messages	Notifications
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	stg_drp_yes_count bigint	target_drp_yes_count bigint
1	51	51

5. Do transaction types reconcile correctly following ETL normalization?

46	CASE
47	WHEN s.txn_type = 'BUY'
48	AND t.transaction_type = 'B'
49	THEN 'PASS'
50	
51	WHEN s.txn_type = 'SELL'
52	AND t.transaction_type = 'S'
53	THEN 'PASS'
54	
55	WHEN s.txn_type = 'ISSUE'
56	AND t.transaction_type = 'I'
57	THEN 'PASS'
58	
59	WHEN s.txn_type = 'DRP'
60	AND t.transaction_type = 'D'
61	THEN 'PASS'

Data Output Messages Notifications						
Showing rows: 1 to 128 Page No: 1 of 1						
	stg_description character varying (100)	target_description character varying (100)	description_validation text	stg_transaction_type character varying (20)	target_transaction_type character varying (10)	transaction_type_validation text
27	MARKET_TRANSFER	MARKET_TRANSFER	PASS	SELL	S	PASS
28	MARKET_TRANSFER	MARKET_TRANSFER	PASS	BUY	B	PASS
29	MARKET_TRANSFER	MARKET_TRANSFER	PASS	SELL	S	PASS
30	MARKET_TRANSFER	MARKET_TRANSFER	PASS	BUY	B	PASS

6. Do transaction counts and aggregate unit balances reconcile between staging and production ledgers?

76	--Aggregate validation
77	SELECT
78	(SELECT COUNT(*) FROM stg.transactions) AS stg_transaction_count,
79	
80	(SELECT COUNT(*) FROM target.share_registry_transactions)
81	AS target_transaction_count,
82	
83	(SELECT SUM(units) FROM stg.transactions)
84	AS stg_total_units,
85	
86	(SELECT SUM(units)
87	FROM target.share_registry_transactions)
88	AS target_total_units;

Data Output Messages Notifications				
Showing rows: 1 to 1				
	stg_transaction_count bigint	target_transaction_count bigint	stg_total_units numeric	target_total_units numeric
1	128	128	1535000.0000	1535000.0000

5.3 Reconstructed Holdings & Final Reconciliation

Following ETL execution, shareholder holdings were reconstructed from the production transaction ledger.

The reconstructed holdings process validated that the production transaction ledger could reproduce the client-provided downstream holdings snapshot following processing of:

- historical registry transactions
- DRP unit generation
- stock split regeneration

5.3.1 Reconstructed Holdings View

Sample reconstruction logic:

```
1 CREATE OR REPLACE VIEW target.vw_reconstructed_holdings AS
2
3 SELECT
4     101 AS company_id,
5
6     t.shareholder_id,
7
8     'AU000000XYZ1' AS identifier,
9
10    SUM(t.units) AS units,
11
12    DATE '2025-12-31' AS as_of_date
13
14 FROM target.share_registry_transactions t
15
16 GROUP BY
17     t.shareholder_id;
```

5.3.2 Holdings Reconciliation

Final downstream holdings were reconciled between:

- client-provided holdings snapshot
- reconstructed production holdings view

Validation objectives included:

Objective	Validation Status
Do reconstructed as-of dates reconcile correctly?	PASS
Does total reconstructed variance reconcile to zero?	PASS

5.3.3 Row-Level Reconciliation Evidence

Row-level reconciliation demonstrating successful downstream holdings reconstruction and zero variance across shareholder positions.

```
24      CASE
25          WHEN v.units = s.units
26              THEN 'PASS'
27              ELSE 'FAIL'
28      END AS units_validation,
29
30      s.as_of_date                AS stg_as_of_date,
31      v.as_of_date                AS reconstructed_as_of_date,
32
33      CASE
34          WHEN s.as_of_date = v.as_of_date
35              THEN 'PASS'
36              ELSE 'FAIL'
37      END AS as_of_date_validation
```

	cted_identifier	stg_units numeric (20,4)	reconstructed_units numeric	variance numeric	units_validation text	stg_as_of_date date	reconstructed_as_of_date date	as_of_date_validation text
1	0XYZ1	506525.0000	506525.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS
2	0XYZ1	358695.0000	358695.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS
3	0XYZ1	48750.0000	48750.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS
4	0XYZ1	18750.0000	18750.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS
5	0XYZ1	12000.0000	12000.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS
6	0XYZ1	3750.0000	3750.0000	0.0000	PASS	2025-12-31	2025-12-31	PASS

5.3.4 Aggregate Reconciliation Summary

Aggregate reconciliation confirming downstream holdings consistency across all shareholder positions.

```
49      --Summary validation
50      SELECT
51          COUNT(*) AS total_positions,
52
53          SUM(
54              CASE
55                  WHEN v.units = s.units
56                      THEN 1
57                      ELSE 0
58              END
59          ) AS matched_positions,
60
61          SUM(v.units - s.units) AS total_variance
62
63      FROM stg.holdings_snapshot s
```

	total_positions bigint	matched_positions bigint	total_variance numeric
1	31	31	0.0000

5.3.5 Reconciliation Outcome

Final reconciliation confirmed:

- successful downstream holdings reconstruction
- successful DRP and stock split processing outcomes
- zero aggregate downstream variance
- reconciliation alignment between reconstructed production holdings and the client source-of-truth holdings snapshot

5.3.6 Deterministic Execution Validation

The migration execution workflow was designed to support rerunnable execution cycles.

Prior to each execution cycle, production target tables were truncated and identity sequences reset to ensure controlled idempotent processing conditions.

Execution summary controls were captured following each migration cycle to validate:

- ETL execution consistency
- production reconciliation consistency
- holdings reconstruction consistency
- aggregate unit consistency
- downstream reconciliation consistency
- zero reconstructed variance

Deterministic replay execution summary:

```
11
12  TRUNCATE TABLE
13      target.share_registry_transactions,
14      target.shareholder_ca_elections,
15      target.stock_split_events,
16      target.dividend_events,
17      target.shareholder_master,
18      target.company_master
19  RESTART IDENTITY CASCADE;
```

Execution summary log:

```
27  INSERT INTO recon.etl_run_summary (
28      run_id,
29      target_company_rows,
30      target_holder_rows,
31      target_txn_rows,
32      total_units,
33      position_variance,
34      status
35  )
36  SELECT
37      1 AS run_id,
38      (SELECT COUNT(*) FROM target.company_master),
39      (SELECT COUNT(*) FROM target.shareholder_master),
40      (SELECT COUNT(*) FROM target.share_registry_transactions),
41      (SELECT SUM(units) FROM target.share_registry_transactions),
42      (
43          SELECT COALESCE(SUM(v.units - s.units),0)
44          FROM stg.holdings_snapshot s
45          LEFT JOIN target.shareholder_master sh
46              ON s.holder = sh.shareholder_name
47          LEFT JOIN target.vw_reconstructed_holdings v
48              ON sh.shareholder_id = v.shareholder_id
49      ),
50      'PASS';
```

Repeated deterministic execution produced fully consistent reconciliation outcomes across all validation layers, confirming reproducible migration behaviour and stable downstream holdings reconstruction.

	id	run_timestamp	target_company_rows	target_holder_rows	target_txn_rows	total_units	position_variance	status
		timestamp without time zone	integer	integer	integer	numeric (20,4)	numeric (20,4)	character varying (20)
1	1	2026-05-10 14:41:10.560351	1	100	128	1535000.0000	0.0000	PASS
2	2	2026-05-10 14:44:25.124368	1	100	128	1535000.0000	0.0000	PASS
3	3	2026-05-10 14:47:14.070529	1	100	128	1535000.0000	0.0000	PASS